IGCSE 0607 Practice Questions — E3.5 — Combined Questions & Markscheme

Asymptotes

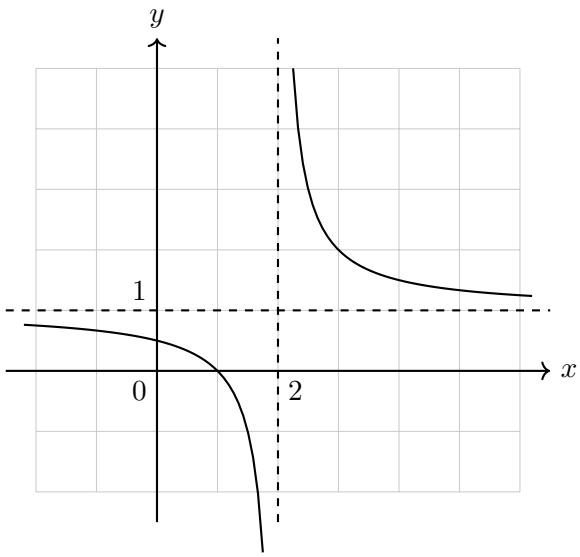
Total Questions: 18 — **Total Marks:** 101 Marks

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  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

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1 The diagram shows the sketch of the graph of a function $y = f(x)$.
The dashed lines are the asymptotes to the curve.



(a) Write down the equation of the vertical asymptote.

..... **[1]**

(b) Write down the equation of the horizontal asymptote.

..... **[1]**

2 Let $f(x) = \dfrac{7}{x-4} + 3$.

(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

3 The function f is given by $f(x) = \dfrac{5}{x+3} - 2$.

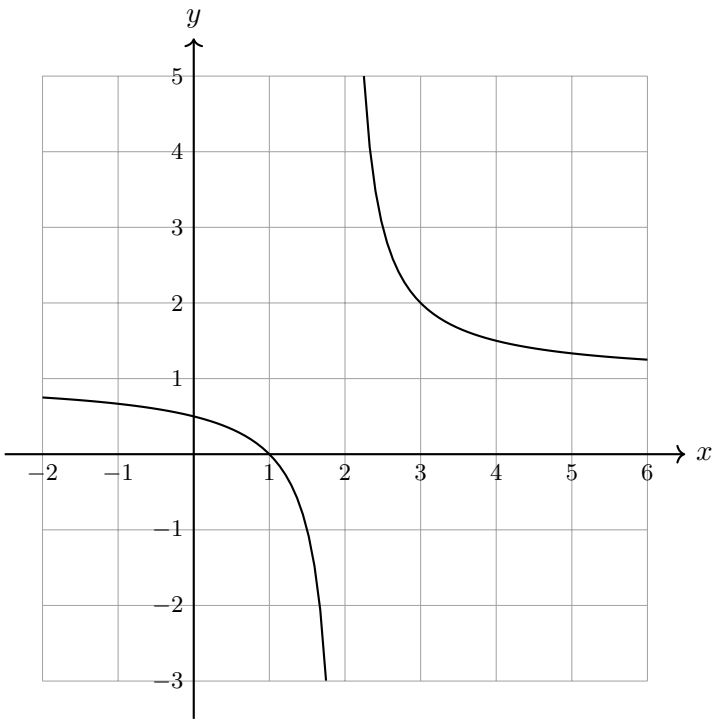
(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

$x =$ ..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

$y =$ ..... **[1]**

4 The diagram shows the graph of a function $y = f(x)$.
The graph has two asymptotes, each parallel to one of the coordinate axes.
The equations of the asymptotes involve only integer values.



(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

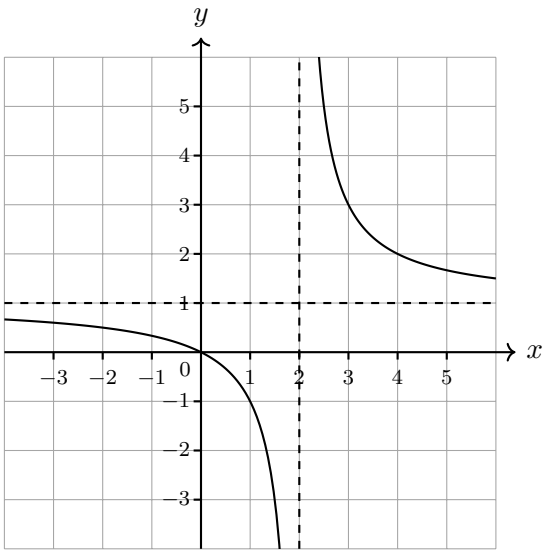
(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

(c) The graph of $y = f(x)$ is reflected in the y-axis to form the graph of $y = g(x)$.
Write down the equation of the vertical asymptote of the graph of $y = g(x)$.

..... **[1]**

5 The diagram shows the graph of a rational function $y = f(x)$.



(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

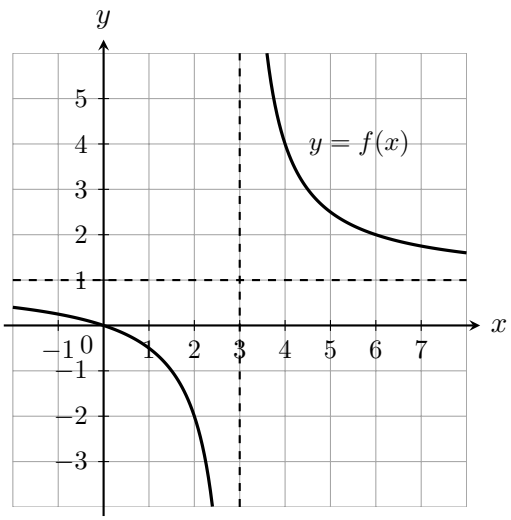
(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

(c) Use the graph to find the value of $f(0)$ and the solution to the equation $f(x) = 0$.

$f(0) =$ ..... , $x =$ ..... **[2]**

6 The diagram shows the graph of $y = f(x)$ and its two asymptotes, which are shown as dashed lines.



(a) Write down the equation of the vertical asymptote of the graph.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph.

..... **[1]**

(c) Use the graph to write down

(i) the value of $f(4)$,

$f(4) =$ ..... **[1]**

(ii) the value of x for which $f(x) = -2$.

$x =$ ..... **[1]**

7 A function is defined as $f(x) = \dfrac{4}{x-1} + 3$.

(a) Write down the equation of the

(i) vertical asymptote,

..... **[1]**

(ii) horizontal asymptote.

..... **[1]**

(b) Find the coordinates of the point where the graph of $y = f(x)$ crosses the x-axis.

((..... , ...)) **[2]**

(c) Find the coordinates of the point where the graph of $y = f(x)$ crosses the y-axis.

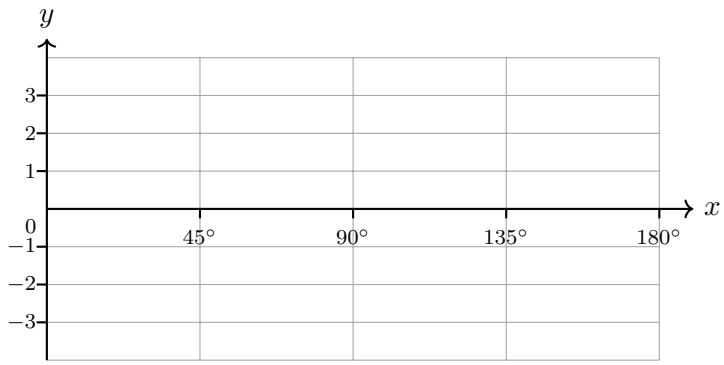
((..... , ...)) **[1]**

8 Consider the trigonometric function $g(x) = \tan(2x)$ for $0° \leq x \leq 180°$.

(a) Write down the equations of the two vertical asymptotes of the graph of $y = g(x)$ for $0° \leq x \leq 180°$.

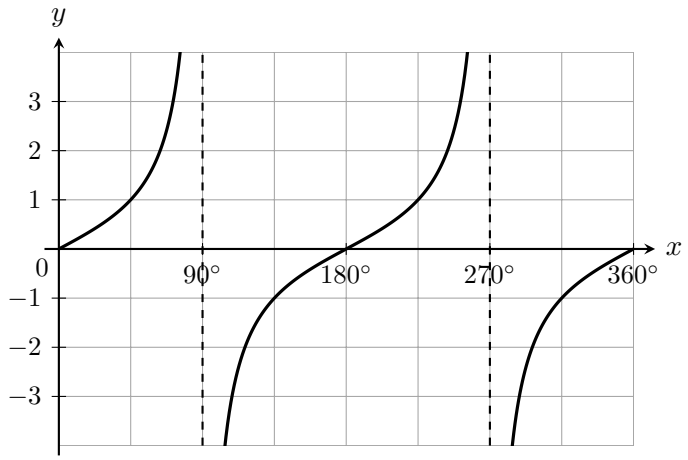
Asymptote 1: ..... , Asymptote 2: ..... **[2]**

(b) On the grid, sketch the graph of $y = \tan(2x)$ for $0° \leq x \leq 180°$, drawing the asymptotes as dashed lines.



[3]

9 The diagram shows the graph of $f(x) = \tan x$ for $0° \leq x \leq 360°$.



(a) Write down the equations of the two asymptotes of the graph of $f(x) = \tan x$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

(b) Explain how the graph shows that $\tan 90°$ is undefined.

..... **[1]**

(c) The graph of $y = \tan x$ is translated by the vector $\begin{pmatrix} 35° \\ 0 \end{pmatrix}$.
Write down the equations of the two asymptotes of the translated graph for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

(d) Given that $\tan 45° = 1$, use the graph to write down the other solution to $\tan x = 1$ for $0° \leq x \leq 360°$.

$x =$ ..... ° **[1]**

10 Consider the functions $g(x) = \tan x$ and $h(x) = \dfrac{4x-1}{2x+5}$.

(a) Write down the equations of the two asymptotes of the graph of $y = g(x)$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

(b) Write down the equations of the vertical and horizontal asymptotes of the graph of $y = h(x)$.

Vertical asymptote $x =$ ..... , Horizontal asymptote $y =$ ..... **[2]**

(c) Solve the equation

$h(x) = 0$

$x =$ ..... **[2]**

11 A function is defined as $f(x) = \dfrac{3}{x+2} + 1$.

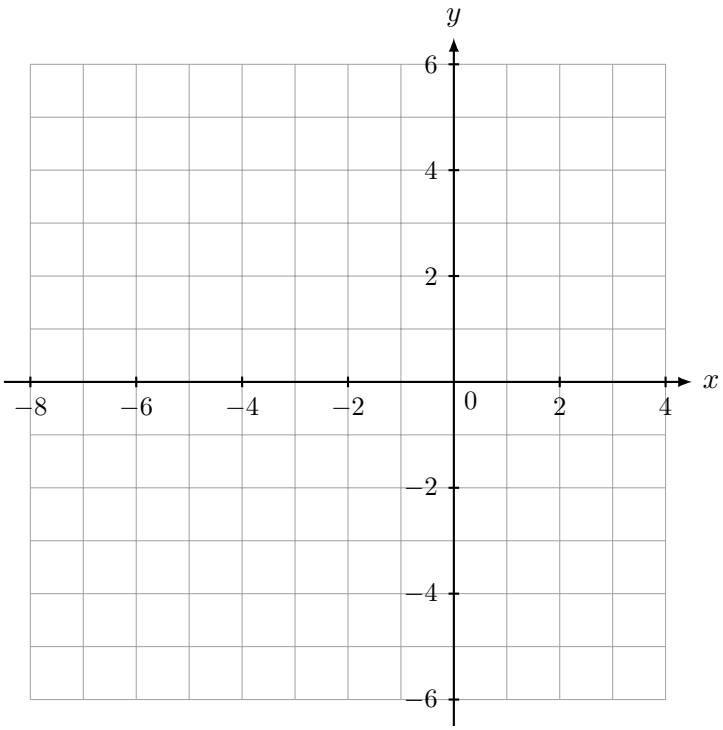
(a) Write down the equations of the two asymptotes of the graph of $y = f(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

(b) Find the coordinates of the points where the graph of $y = f(x)$ crosses the x-axis and the y-axis.

x-axis at ..... , y-axis at ..... **[2]**

(c) On the grid, sketch the graph of $y = f(x)$.



[3]

12 Let $f(x) = \dfrac{3}{x+2} - 4$.

(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

(c) Find the coordinates of the points where the graph of $y = f(x)$ intersects the coordinate axes.

x-intercept = ... , y-intercept = ... **[5]**

(d) The graph of $y = \tan x$ has an infinite number of vertical asymptotes.
Write down the equations of the two asymptotes in the interval $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

13 A function is given by $f(x) = \dfrac{a}{x-b} + c$, where a, b and c are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = -1$.
The graph intersects the x-axis at the point $(5, 0)$.

(a) Write down the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[2]**

(b) Work out the value of a.

$a =$ ..... **[2]**

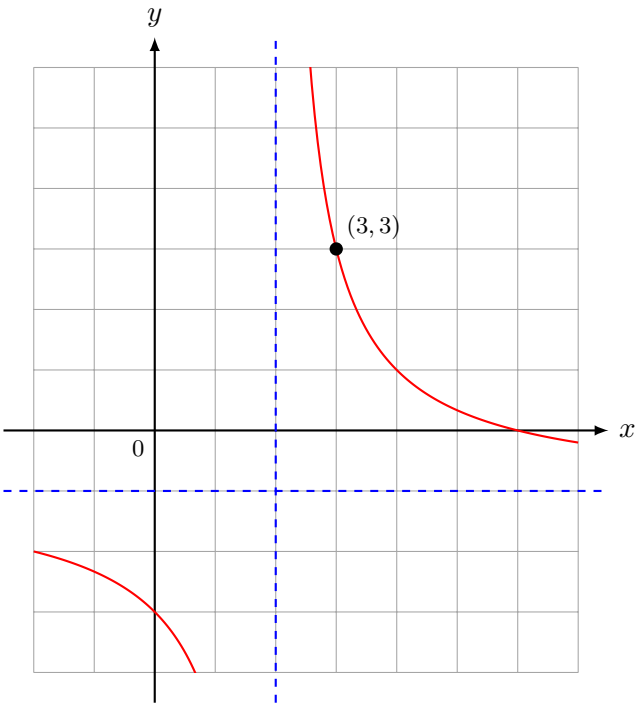
(c) The graph of $y = f(x)$ intersects the straight line $y = -x + 5$ at two points.
Find the x-coordinates of these two points.

$x =$ ..... , $x =$ ..... **[2]**

14 The diagram shows the graph of $y = f(x)$ where $f(x) = \dfrac{p}{x-q} + r$.

The graph has asymptotes at $x = 2$ and $y = -1$.

The graph passes through the point $(3, 3)$.



(a) Find the values of q and r.

$q =$ ..... , $r =$ ..... **[2]**

(b) Find the value of p.

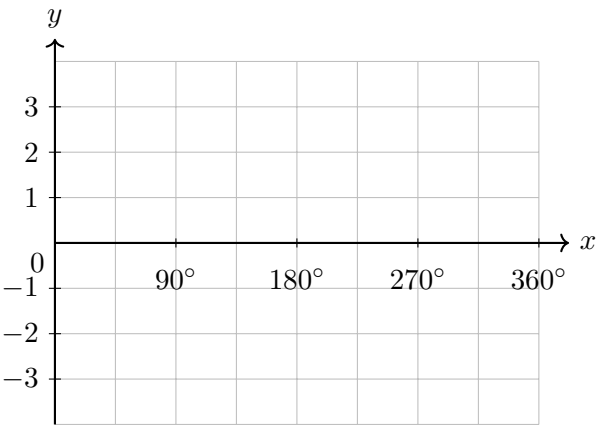
$p =$ ..... **[2]**

(c) Find the coordinates of the point where the graph of $y = f(x)$ crosses the x-axis.

(..... , ...) **[3]**

15 The function $g(x) = \tan x$ is defined for $0° \leq x \leq 360°$.

(a) On the grid, sketch the graph of $y = g(x)$.



[2]

(b) Write down the equations of the asymptotes of the graph of $y = g(x)$ in this domain.

[2]

(c) Solve the equation $2\tan x + 3 = 0$ for $0° \leq x \leq 360°$.
Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

16 Consider the function $h(x) = \dfrac{2x^2 - 5x - 3}{x - 2}$ for $-4 \leq x \leq 6$, $xe2$.

(a) Use your graphic display calculator to find the zeros of $h(x)$.

$x =$ ..... , $x =$ ..... **[2]**

(b) Write down the equation of the vertical asymptote of the graph of $y = h(x)$.

..... **[1]**

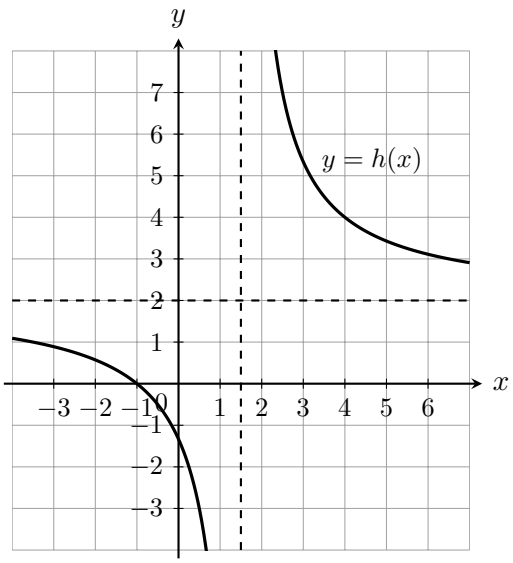
(c) Explain why the graph of $y = h(x)$ does not have a horizontal asymptote.

..... **[2]**

(d) The equation $h(x) = k$ has no real solutions.
Find the range of possible values of k, giving your boundary values correct to 2 decimal places.

..... **[2]**

17 The diagram shows the graph of $y = h(x)$, where $h(x) = \dfrac{5}{x - 1.5} + 2$ for $xe1.5$.
The dashed lines show the asymptotes of the curve.



(a) From the graph, write down the equation of

(i) the vertical asymptote,

..... **[1]**

(ii) the horizontal asymptote.

..... **[1]**

(b) Explain why the equation $h(x) = 2$ has no solution.

..... **[1]**

(c) The graph of $y = h(x)$ is reflected in the y-axis.
Write down the equations of the vertical and horizontal asymptotes of the reflected graph.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

(d) Solve the equation $h(x) = 3.4$, giving your answer correct to 3 significant figures.

$x =$ ..... **[3]**

18 The function $f(x)$ is given by $f(x) = \dfrac{p}{x-q} + r$, where p, q, and r are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 5$ and a horizontal asymptote at $y = -3$.
The graph passes through the point $(7, -2)$.

(a) Write down the value of q and the value of r.

$q =$ ..... , $r =$ ..... **[2]**

(b) Find the value of p.

$p =$ ..... **[3]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

(d) Write down the equations of the asymptotes of the graph of $y = f^{-1}(x)$.

$x =$ ..... , $y =$ ..... **[2]**

1 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

[Back to Question] *[Total Marks: 2]*

2 **(a)** $x = 4$ **[B1]**

(b) $y = 3$ **[B1]**

[Back to Question] *[Total Marks: 2]*

3 **(a)** -3 **[B1]**

(b) -2 **[B1]**

[Back to Question] *[Total Marks: 2]*

4 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

(c) $x = -2$ **[B1]**

[Back to Question] *[Total Marks: 3]*

5 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

(c) $f(0) = 0$ **[B1]**

$x = 0$ **[B1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** $x = 3$ **[B1]**

(b) $y = 1$ **[B1]**

(c) **(i)** 4 **[B1]**

(ii) 2 **[B1]**

[Back to Question] *[Total Marks: 4]*

7 **(a)** **(i)** $x = 1$ **[B1]**

(ii) $y = 3$ **[B1]**

(b) $\dfrac{4}{x-1} + 3 = 0$ **[M1]**

$(-\frac{1}{3}, 0)$ **[A1]**

(c) $(0, -1)$ **[B1]**

[Back to Question] *[Total Marks: 5]*

8 (a) $x = 45°$ **[B1]**

$x = 135°$ **[B1]**

(b) Draws dashed vertical lines at $x = 45°$ and $x = 135°$ **[B1]**

Correct tangent curve shape passing through $(0°, 0)$, $(90°, 0)$, and $(180°, 0)$ **[B1]**

Branches approach asymptotes cleanly without touching or curling back **[B1]**

[Back to Question] *[Total Marks: 5]*

9 (a) $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

(b) There is a vertical asymptote at $x = 90°$ (or the curve never touches $x = 90°$) **[B1]**

(c) $x = 125°$ **[B1]**

$x = 305°$ **[B1]**

(d) 225 **[B1]**

[Back to Question] *[Total Marks: 6]*

10 (a) $90°$ **[B1]**

$270°$ **[B1]**

(b) Vertical asymptote: $x = -2.5$ (or $-\frac{5}{2}$) **[B1]**

Horizontal asymptote: $y = 2$ **[B1]**

(c) $4x - 1 = 0$ **[M1]**

0.25 (or $\frac{1}{4}$) **[A1]**

[Back to Question] *[Total Marks: 6]*

11 (a) Vertical asymptote: $x = -2$ **[B1]**

Horizontal asymptote: $y = 1$ **[B1]**

(b) Substitute $y = 0$ to find x-intercept: $0 = \frac{3}{x+2} + 1 \implies x = -5$ **[M1]**

x-axis at $(-5, 0)$ and y-axis at $(0, 2.5)$ **[A1]**

(c) Two correct branches of the hyperbola sketched **[B1]**

Graph approaches $x = -2$ and $y = 1$ correctly without touching or crossing them **[B1]**

Graph passes through $(-5, 0)$ and $(0, 2.5)$ approximately **[B1]**

[Back to Question] *[Total Marks: 7]*

12 (a) $x = -2$ **[B1]**

(b) $y = -4$ **[B1]**

(c) Sets $x = 0$ to find the y-intercept: $y = \frac{3}{0+2} - 4$ **[M1]**

y-intercept is $(0, -2.5)$ **[A1]**

Sets $y = 0$ to find the x-intercept: $0 = \frac{3}{x+2} - 4$ **[M1]**

$4 = \frac{3}{x+2} \implies 4(x+2) = 3 \implies 4x + 8 = 3$ **[M1]**

x-intercept is $(-1.25, 0)$ **[A1]**

(d) $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

[Back to Question] *[Total Marks: 9]*

13 (a) $b = 2$ **[B1]**

$c = -1$ **[B1]**

(b) Substitutes $(5, 0)$ into the equation: $\frac{a}{5-2} - 1 = 0$ **[M1]**

$a = 3$ **[A1]**

(c) Sets up the equation $\frac{3}{x-2} - 1 = -x + 5$ or uses GDC to find intersection of $y = f(x)$ and $y = -x + 5$ **[M1]**

$x = 3$ and $x = 5$ **[A1]**

[Back to Question] *[Total Marks: 6]*

14 (a) $q = 2$ **[B1]**

$r = -1$ **[B1]**

(b) Substitutes $(3, 3)$, $q = 2$ and $r = -1$ into $f(x)$: $3 = \frac{p}{3-2} - 1$ **[M1]**

$p = 4$ **[A1]**

(c) Sets $f(x) = 0$

$\frac{4}{x-2} - 1 = 0$ [M1]

$$\frac{4}{x-2} = 1$$

x - 2 = 4[M1]

$(6, 0)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

15 (a) Two correct vertical asymptotes at $x = 90°$ and $x = 270°$ **[B1]**

Three correct branches passing through $(0, 0)$, $(180, 0)$, and $(360, 0)$ with correct general curvature **[B1]**

(b) $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

(c) $\tan x = -1.5$ **[M1]**

$x = 123.7°$ **[A1]**

$x = 303.7°$ **[A1]**

[Back to Question] *[Total Marks: 7]*

16 (a) $x = -0.5$ (or $-\frac{1}{2}$) **[B1]**

$x = 3$ **[B1]**

(b) $x = 2$ **[B1]**

(c) As x

$\pm\infty$, $h(x)$ increases or decreases without bound (does not approach a constant value) [B1]

Mentions that the degree of the numerator (2) is greater than the degree of the denominator (1), or that the curve approaches an oblique/sloping line $y = 2x - 1$ **[B1]**

(d) Finds local maximum at $(0.27, 4.07)$ and local minimum at $(3.73, 1.93)$ using GDC **[M1]**

$1.93 < k < 4.07$ **[A1]**

[Back to Question] *[Total Marks: 7]*

17 (a) (i) $x = 1.5$ **[B1]**

(ii) $y = 2$ **[B1]**

(b) $y = 2$ is a horizontal asymptote so the curve never reaches $y = 2$ **[B1]**

(c) Vertical asymptote: $x = -1.5$ **[B1]**

Horizontal asymptote: $y = 2$ **[B1]**

(d) $\frac{5}{x - 1.5} + 2 = 3.4$ or $\frac{5}{x - 1.5} = 1.4$ **[M1]**

$x - 1.5 = \frac{5}{1.4}$ **[M1]**

5.07 **[A1]**

[Back to Question] *[Total Marks: 8]*

18 **(a)** $q = 5$ **[B1]**

$r = -3$ **[B1]**

(b) Substitutes $x = 7, y = -2$ and their values of q and r into the equation: $-2 = \frac{p}{7-5} - 3$ **[M1]**

$-2 = \frac{p}{2} - 3 \implies \frac{p}{2} = 1$ **[M1]**

$p = 2$ **[A1]**

(c) Sets $y = \frac{2}{x-5} - 3$ and attempts to make x the subject, or swaps x and y **[M1]**

$y + 3 = \frac{2}{x-5} \implies x - 5 = \frac{2}{y+3}$ **[M1]**

$x = \frac{2}{y+3} + 5$ **[M1]**

$f^{-1}(x) = \frac{2}{x+3} + 5$ or equivalent **[A1]**

(d) $x = -3$ **[B1]**

$y = 5$ **[B1]**

(Strict follow through from their answers in part (a): $x = r, y = q$)

[Back to Question] *[Total Marks: 11]*